

1.8: Differential Equations

1. Terminology and Classification

A differential equation (DE) relates a function to its derivatives. The **order** is the highest derivative. A DE is **linear** if the unknown and its derivatives appear only to the first power and not multiplied together. Linear DEs are **homogeneous** if the RHS is zero.

OCR A covers first-order linear DEs, second-order linear DEs with constant coefficients, and coupled first-order systems.

2. First-Order Linear DEs — Integrating Factor

For a DE of the form

$$\frac{dy}{dx} + P(x)y = Q(x),$$

multiply through by the **integrating factor**

$$\mu(x) = e^{\int P(x) dx}.$$

The left side becomes $\frac{d}{dx}(\mu y)$, so

$$\mu y = \int \mu(x) Q(x) dx + C.$$

Solve $\frac{dy}{dx} + \frac{2y}{x} = x$.

Here $P = 2/x$, so $\mu = e^{\int 2/x dx} = e^{2 \ln x} = x^2$.

Multiply: $x^2 y' + 2xy = x^3 \Rightarrow \frac{d}{dx}(x^2 y) = x^3$.

Integrate: $x^2 y = \frac{x^4}{4} + C \Rightarrow y = \frac{x^2}{4} + \frac{C}{x^2}$.

3. Second-Order Linear Homogeneous DEs with Constant Coefficients

For

$$a \frac{d^2 y}{dx^2} + b \frac{dy}{dx} + cy = 0,$$

try $y = e^{mx}$ to obtain the **auxiliary equation** $am^2 + bm + c = 0$.

Roots	General solution
Two distinct real m_1, m_2	$y = Ae^{m_1 x} + Be^{m_2 x}$
Repeated real m	$y = (A + Bx)e^{mx}$
Complex $m = p \pm iq$	$y = e^{px}(A \cos qx + B \sin qx)$

Solve $y'' - 4y' + 13y = 0$.

Auxiliary: $m^2 - 4m + 13 = 0 \Rightarrow m = 2 \pm 3i$.

GS: $y = e^{2x}(A \cos 3x + B \sin 3x)$.

4. Non-Homogeneous Second-Order DEs

For $ay'' + by' + cy = f(x)$ the general solution is

$$y = y_{CF} + y_{PI}$$

- y_{CF} (complementary function): the GS of the *homogeneous* version (§3).
- y_{PI} (particular integral): any single function satisfying the full DE.

Trial Functions for PI

$f(x)$	Trial y_{PI}
Polynomial of degree n	General polynomial of degree n
$e^{\lambda x}$	$Pe^{\lambda x}$ (unless λ is already a root of the aux eqn)
$\sin \omega x$ or $\cos \omega x$	$P \cos \omega x + Q \sin \omega x$
Combination	Sum of the corresponding trials

If the chosen trial form is already part of y_{CF} , multiply by x (or x^2 for a repeated root) before substituting.

Solve $y'' - 3y' + 2y = 4e^{3x}$.

CF: $m^2 - 3m + 2 = 0 \Rightarrow m = 1, 2$. $y_{CF} = Ae^x + Be^{2x}$.

PI: try $y = Pe^{3x}$. $y' = 3Pe^{3x}$, $y'' = 9Pe^{3x}$.

Substitute: $(9P - 9P + 2P)e^{3x} = 4e^{3x} \Rightarrow 2P = 4 \Rightarrow P = 2$.

GS: $y = Ae^x + Be^{2x} + 2e^{3x}$.

Boundary/Initial Conditions

Use to pin down A, B . Apply after writing the GS.

5. Coupled First-Order Linear Systems

A pair like

$$\frac{dx}{dt} = 3x + y, \quad \frac{dy}{dt} = -2x + y$$

can be solved by **elimination**: differentiate one equation and substitute to get a second-order DE in a single variable.

From the system above:

Differentiate the first: $x'' = 3x' + y'$. Substitute $y' = -2x + y$ and $y = x' - 3x$:

$$x'' = 3x' + (-2x) + (x' - 3x) = 4x' - 5x.$$

So $x'' - 4x' + 5x = 0$. Aux: $m^2 - 4m + 5 = 0 \Rightarrow m = 2 \pm i$.

$$x = e^{2t}(A \cos t + B \sin t), \text{ and } y = x' - 3x.$$

Alternatively (richer option — FP2 level): write as a matrix system $\dot{\mathbf{z}} = \mathbf{A}\mathbf{z}$ and diagonalise (see §1.2).

6. Modelling with Differential Equations

Newton's Law of Cooling

$$\frac{dT}{dt} = -k(T - T_{\text{env}})$$

Separable: $T = T_{\text{env}} + Ce^{-kt}$.

Exponential Growth/Decay

$$\frac{dN}{dt} = kN \Rightarrow N = N_0 e^{kt}$$

Mechanical Oscillators (Damped Harmonic Motion)

$$m\ddot{x} + c\dot{x} + kx = F(t)$$

- **Underdamped** ($c^2 < 4mk$): complex roots — oscillation with decaying amplitude.
- **Critically damped** ($c^2 = 4mk$): repeated real root — fastest non-oscillating decay.
- **Overdamped** ($c^2 > 4mk$): two real negative roots — slow decay without oscillation.

Resonance

When the driving frequency matches the natural frequency of an undamped oscillator, the PI trial $P \cos \omega t + Q \sin \omega t$ fails because it's already in y_{CF} . Multiplying by t gives unbounded growth — *resonance*.

7. Exam Style Strategy

- First-order linear: identify $P(x)$, compute μ , *don't forget* the constant of integration.
- Second-order: always write the auxiliary equation and classify its roots before the PI trial.
- If your PI trial clashes with the CF, multiply by x — this is the single most common mistake.
- Always apply initial conditions *after* writing the GS, and simplify.
- In modelling questions, interpret constants physically (e.g. " $k > 0$ represents the cooling rate").